

Paths contradict the proposed acquaintance bound

Conjecture 00000001679

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Abstract

A path on N vertices exposes only $N - 1$ unordered pairs of agents at each time. Even granting arbitrary rearrangements between times, complete acquaintance requires at least $N/2 - 1$ rounds. For sufficiently large N this exceeds $1024N/\log N$, disproving the stated universal upper bound.

1 The actual acquaintance process

Place one labeled agent at each vertex of the path P_N . Agents become acquainted when their occupied vertices are adjacent. At each round the usual game permits swaps along a matching of graph edges. Count adjacency in the initial configuration as time zero, so r rounds provide $r + 1$ configurations. This convention gives the proposed upper bound its most favorable starting point.

For the lower bound we allow an arbitrary permutation of the agents at every time. This contains every legal matching-swap schedule and can only make acquaintance easier. No assumption about which permutations can be reached in a round is needed.

Theorem 1. *If all N agents on P_N are acquainted after r rounds, where $N \geq 2$, then*

$$N \leq 2(r + 1), \quad r \geq N/2 - 1.$$

Proof. The path has $N - 1$ edges. At each time these expose exactly $N - 1$ unordered pairs, or $2(N - 1)$ ordered pairs of distinct agents. Repeated meetings add no new pairs. The union of the ordered pairs exposed at times $0, \dots, r$ therefore has size at most $2(r + 1)(N - 1)$. Complete acquaintance requires all $N(N - 1)$ ordered pairs of distinct agents. Hence

$$N(N - 1) \leq 2(r + 1)(N - 1).$$

Canceling the positive factor $N - 1$ proves the result. □

2 Contradiction to the conjectured constant

Choose an integer $N \geq 4$ with $\log N > 4096$; for example, any integer larger than $e^{4096} + 5$ will do. Then

$$\frac{1024N}{\log N} < \frac{N}{4} \leq \frac{N}{2} - 1.$$

Thus no schedule completing acquaintance on this connected path can have $r \leq 1024N/\log N$. This contradicts the conjectured bound $a(P_N) \leq 2^{10}N/\log N$. The proof is existential and exact, without approximating e^{4096} or evaluating a graph of that size. The logarithm is natural, as usual; any other fixed base leads to the same asymptotic contradiction.

3 Formal verification

Lean represents the graph by Mathlib’s actual `SimpleGraph.pathGraph (n + 1)`. It proves that these paths are connected. The formal parameter n counts edges, so the paper’s vertex count is $N = n + 1$. A schedule is a function from `Fin (r + 1)` to permutations of `Fin (n + 1)`, assigning each agent its actual position.

The proposition `AllMet` says that each two distinct agents occupy adjacent vertices at some observed time. It is expressed using the path graph’s adjacency relation, rather than an independently invented counting quantity. A finite map `visiblePair` enumerates each time, each path edge, and its two orientations, and then transports the edge’s endpoints back to agents with the inverse position permutation.

- `adjacent_is_visible` proves that every actual meeting is in that finite image.
- `counting_lower_bound` places all ordered distinct-agent pairs inside the image, bounds its cardinality by the domain’s cardinality, and cancels $n > 0$ to obtain $N \leq 2(r + 1)$.
- `no_short_schedule` combines this inequality with $N \geq 4$, $\log N > 4096$, and the conjectured upper bound to obtain a contradiction.
- `exists_bad_path` constructs a sufficiently large natural vertex count through the Archimedean property and uses the exact monotonicity of the real logarithm. It quantifies over every round count satisfying the claimed bound and every position schedule.

The formal theorem is stronger than needed: even schedules with unrestricted rearrangements fail within the claimed number of rounds. Every actual adjacent-swap schedule supplies such a sequence of position permutations. Accordingly the proof does not need to compute the exact optimal acquaintance time or formalize an optimization algorithm.

Scope. The source gives a universal bound in terms of the number of vertices; it imposes no expansion, density, or random-graph hypothesis. The examples are ordinary finite connected paths. This disproof addresses that universal upper bound; it makes no assertion about the best bounds for separately restricted graph families.

Reproduction. In `lean/`, run:

```
lake build
lake env lean -DwarningAsError=true Main.lean
```

The project pins Lean 4.19.0 and an exact Mathlib revision. The final theorems use only standard logical axioms, with no unproved placeholders, custom axioms, native evaluation, or external numerical certificates. Run `tectonic main.tex` to rebuild the PDF. The bilingual source and its immutable provenance are included with the submission.

References

- [1] The Last Math Competition, Conjecture 00000001679, revision 9b795e7a94a6.
- [2] The Mathlib Community, Mathlib, revision c44e0c8ee63c, module `Combinatorics.SimpleGraph.Hasse`.